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Bhuvann Vinodh Ram

Graduate Data Engineer

Stamp 1G holder: eligible to work full-time in Ireland without a work permit

PROFILE

MSc AI for Medicine graduate (UCD) who builds reproducible data pipelines: automated extraction from APIs and bulk files, cleaning and modelling in pandas, tests and CI in GitHub Actions, and scheduled refreshes that publish results to the web. Industry experience containerising applications with Docker and deploying them through GitLab CI/CD. Currently a Trust and Safety Associate at Accenture, Dublin.

EDUCATION

MSc Artificial Intelligence for Medicine and Medical Research — 2024 – 2025
University College Dublin — GPA 3.37 / 4.2

BTech Computer Science and Medical Engineering — 2020 – 2024
Sri Ramachandra Institute of Higher Education and Research, Chennai — CGPA 3.18 / 4

TECHNICAL SKILLS

Languages	Python, SQL, R (basic), Bash
Pipelines	ETL with pandas and NumPy, REST APIs, bulk file ingestion, web scraping (Requests, Beautiful Soup), data validation
Engineering	Git, GitHub Actions (CI and scheduled jobs), GitLab CI/CD, Docker, pytest, Linux
Data stores	CSV and JSON, MySQL, MongoDB (basic)
Also	scikit-learn, Tableau, Jupyter

EXPERIENCE

Trust and Safety Associate — Mar 2026 – Present
Accenture — Dublin, Ireland

- Evaluate, classify and moderate user-generated content against client platform policies to keep online communities safe and compliant.

Research Intern — Apr 2025 – Sep 2025
University College Dublin — Dublin, Ireland

- Integrated TCGA colon adenocarcinoma RNA-seq, copy-number and DNA methylation data for a shared set of patients into one analysis-ready dataset.
- Built a variational autoencoder to compress the three data types into a single latent representation for downstream clustering.

Project Intern — Jan 2024 – Jun 2024
Centre for Toxicology and Development, SRIHER — Chennai, India

- Built a video-to-data pipeline that turned zebrafish behaviour videos into numerical tracking data for drug-testing studies.

Research Intern — Dec 2022 – Feb 2023
IIITDM Kancheepuram — Chennai, India

- Developed and compared CNN breast cancer classifiers using pre-trained VGG16, ResNet, MobileNet and EfficientNet models.

Intern — Feb 2022 – Apr 2022
Smartail Pvt. Ltd. — Chennai, India

- Containerised web applications with Docker and deployed them to servers through GitLab CI/CD pipelines.

PROJECTS

Conflict Risk Monitor 2026
Python · pandas · scikit-learn · GitHub Actions · JavaScript Live demo · GitHub

- End-to-end pipeline: downloads UCDP conflict-event data (final and monthly releases) and World Bank indicators, builds a 176-country monthly panel, and publishes JSON to a static dashboard.
- Scheduled GitHub Actions job refreshes data and retrains every month; pytest suite includes a check that features never use future data.

Tamil Thedal (தமிழ் தேடல்) 2026
JavaScript · OpenCV.js · Tesseract.js · Python ETL Live demo · GitHub

- ETL script builds 1,549 verses from open sources at a pinned commit, keeps only public-domain fields, adds themes, and is rebuilt by GitHub Actions with tests.

Ovarian cancer survival prediction (HGSOC) 2025
XGBoost · SHAP · TCGA GitHub

- XGBoost on TCGA clinical and mutation data with SHAP explanations; found and removed follow-up variables that leaked the outcome, then reran the notebook end to end in GitHub Actions: test accuracy 0.78, ROC AUC 0.91.

Retail sales register analysis 2025
Requests · Beautiful Soup · pandas GitHub

- Scraped multi-page sales records, cleaned dates, currencies and missing values, and analysed revenue, profit and customer behaviour by location and age group.

Brain tumour classification from MRI 2025
CNN · OpenCV · Streamlit · pytest GitHub

- Four-class CNN (0.80 accuracy, 0.77 macro F1 on 2,258 test images) and ResNet50 transfer learning, deployed as a Streamlit app with unit tests and CI.

OTHER PROJECTS

- **ECG heartbeat classification** (1D CNN · Keras, 2025): 1D CNN classifying MIT-BIH heartbeats into five classes: 96.01% test accuracy on 21,892 held-out beats, reproduced by an automated GitHub Actions rerun.
- **Student engagement analysis (VLE)** (scikit-learn · ensembles, 2025): Voting ensembles (random forest, logistic regression, gradient boosting, SVM) on virtual learning environment data, compared across feature-selection settings.
- **RNA-seq analysis of HER2 breast cancer** (R · DESeq2 · clusterProfiler, 2024): Differential expression, pathway enrichment and survival predictors in R to study how HER2 amplification changes gene expression.
- **Wimbledon Championship dashboard** (Tableau, 2021): Interactive Tableau dashboard of Wimbledon results and player statistics.

AWARDS AND ACTIVITIES

- UCD Innovation Challenge 2024: Innovation Spark Award
- NASA Space Apps Challenge 2024: built a climate storytelling website with a team
- UCD Advantage Award
- First place, Tamil poetry competition (Agal Tamil Club)
- Second place, Palaestra intramural volleyball tournament